



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - Jan-Feb

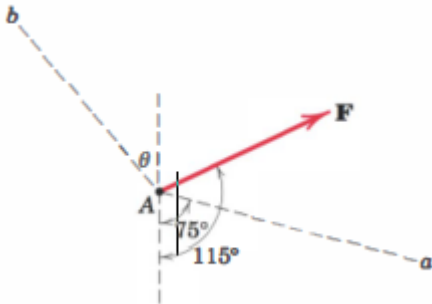
UE24CV141A - Engineering Mechanics- Statics

Total Marks : 100.0

1.a. Explain Transmissibility of a force with a neat sketch. (5.0 Marks)

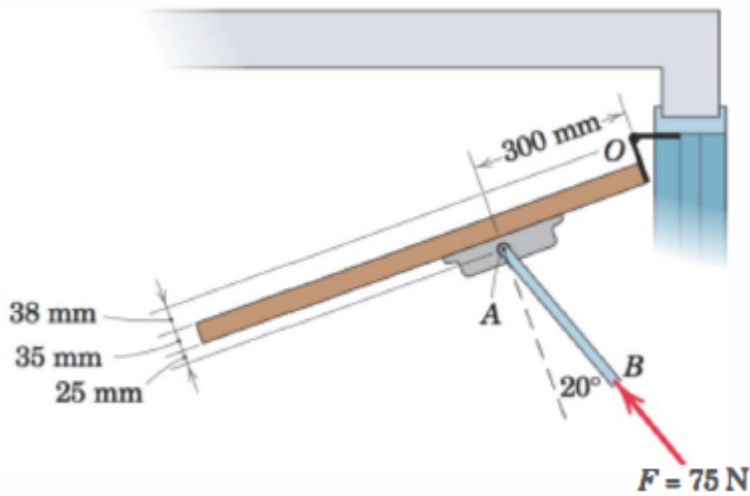
1.b. If the projection P_a and component F_b of the force F along oblique axes a and b are both 325 N, determine the magnitude F and the orientation θ of the b -axis.

(5.0 Marks)



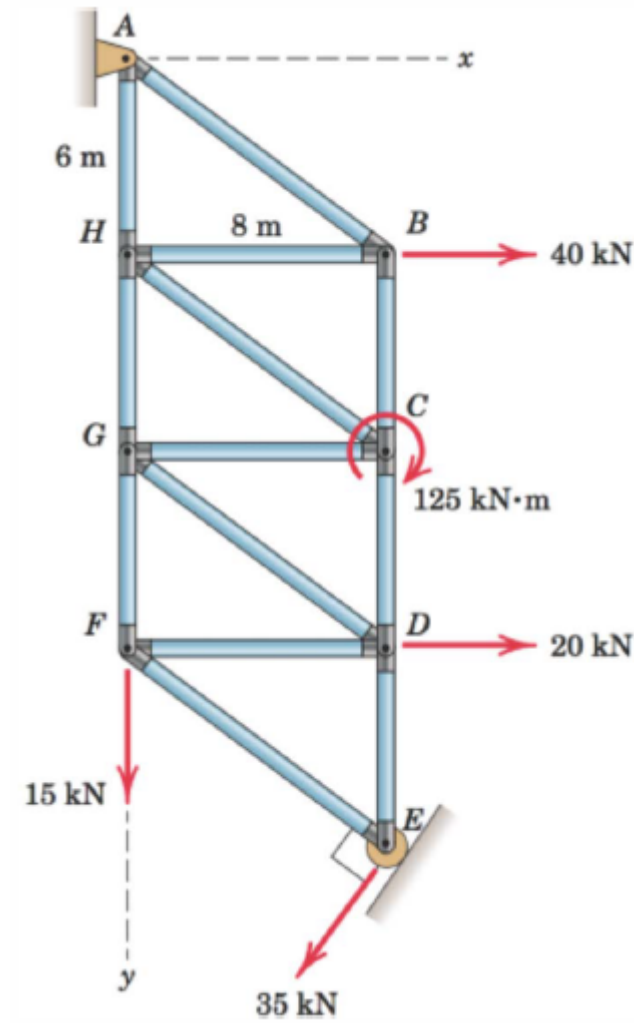
1.c. An overhead view of a door is shown. If the compressive force F acting in the coupler arm of the hydraulic door closer is 75 N with the orientation shown, determine the moment of this force about the hinge axis O .

(5.0 Marks)



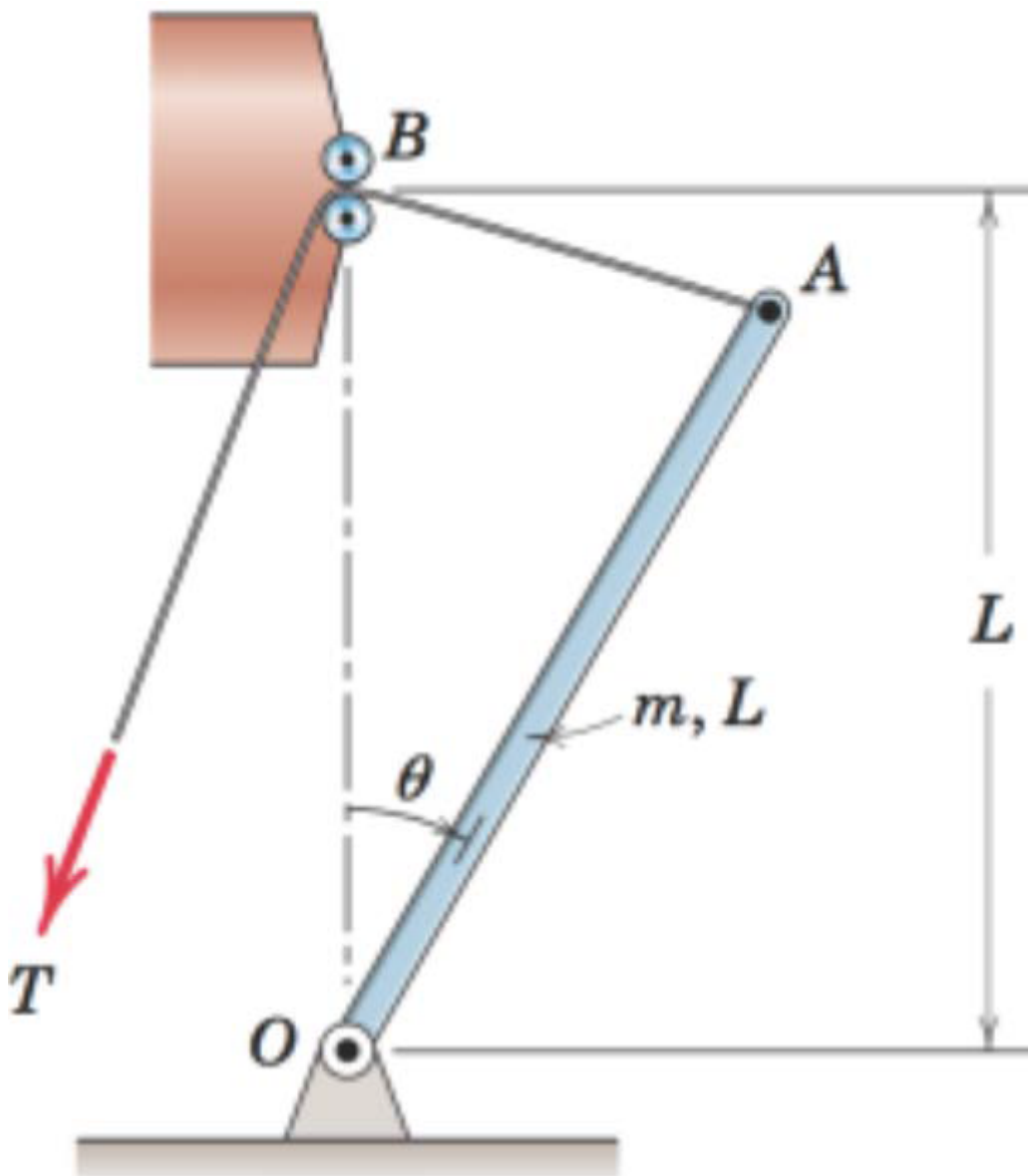
1.d. For the truss loaded as shown, determine the equation for the line of action of the stand-alone resultant R and state the coordinates of the points on the x - and y -axes through which the line of action passes. All triangles are 3-4-5.

(10.0 Marks)



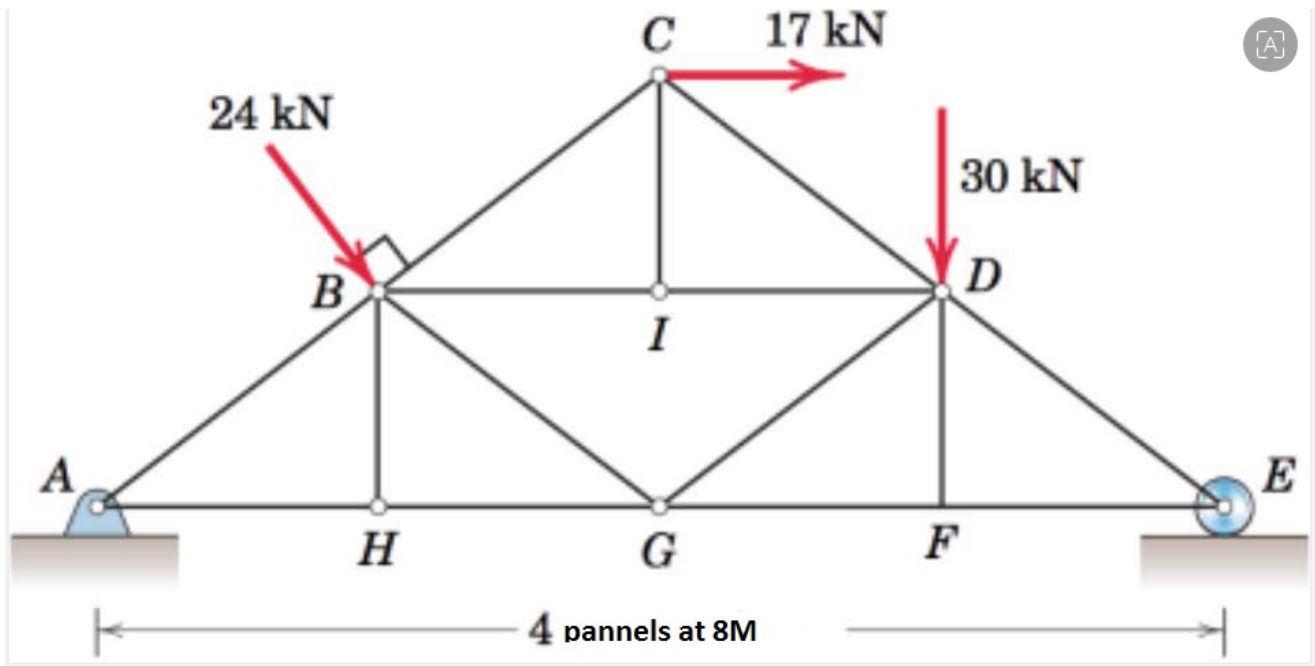
2.a. Determine the force T required to hold the uniform bar of mass m and length L in an arbitrary angular position θ . Plot your result over the range $0 \leq \theta \leq 90^\circ$, and state the value of T for $\theta = 40^\circ$.

(10.0 Marks)



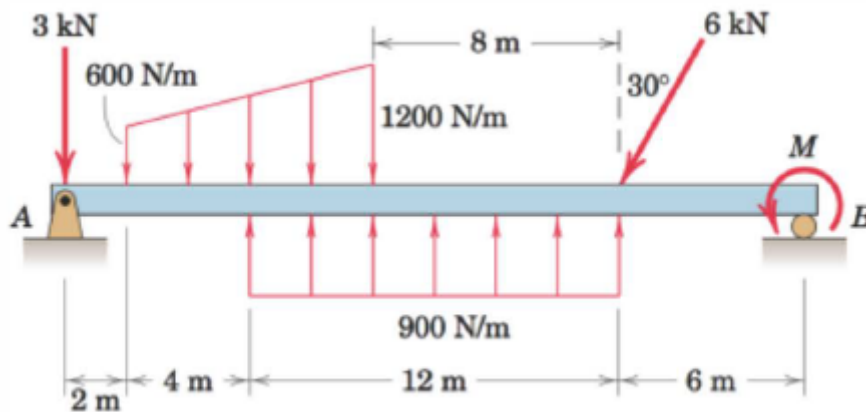
2.b. Explain both internal and external statically determinate, statically indeterminate and Redundancy with respect to trusses. (5.0 Marks)

2.c. Determine the force in each member of the loaded truss. All triangles are 3-4-5.



(10.0 Marks)

3.a. Determine the magnitude of the moment M which will cause the beam to just begin to lift off the roller at B. For this value of M , determine the magnitude of the pin reaction at A.



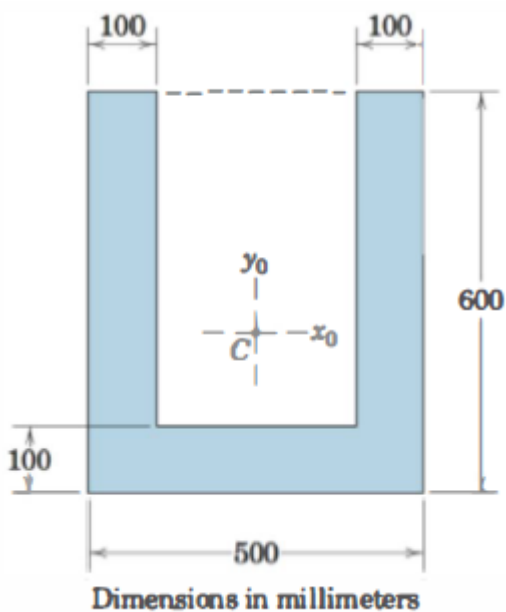
(10.0 Marks)

3.b. Explain with neat sketch Radius of Gyration.

(5.0 Marks)

3.c. Calculate the polar radius of gyration of the shaded area about its centroid C.

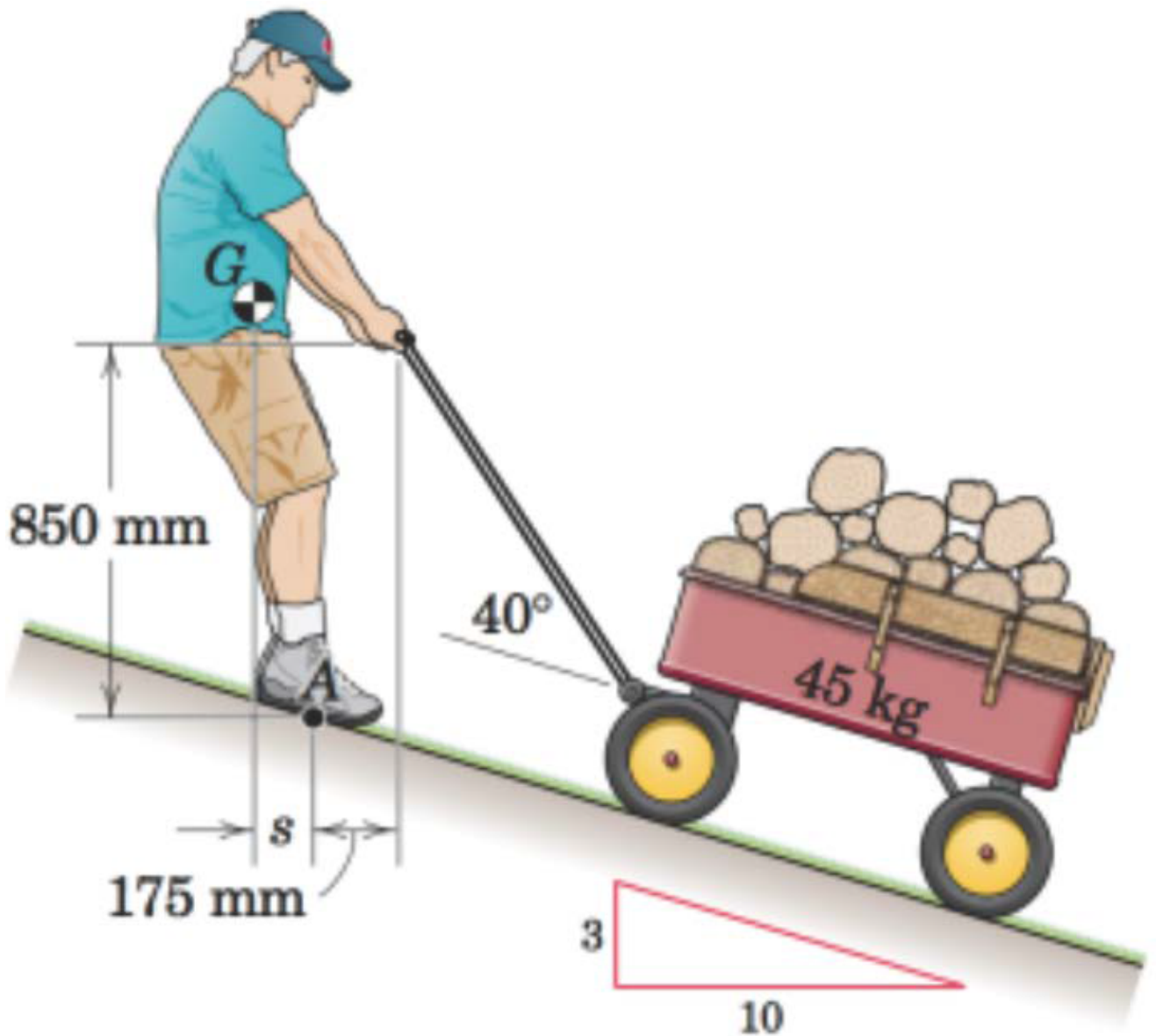
(10.0 Marks)



4.a. Explain the terms Angle of friction and Cone of friction.

(5.0 Marks)

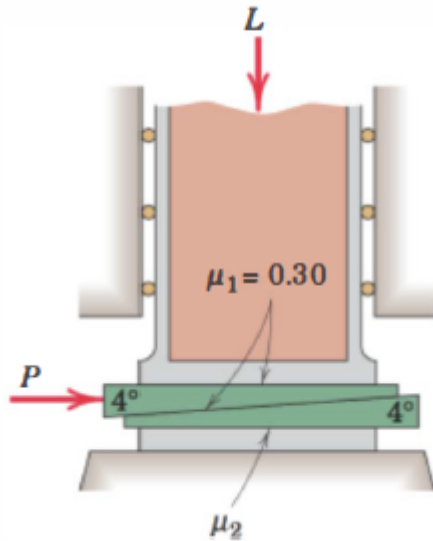
4.b. A 82-kg man pulls the 45-kg cart up the incline at steady speed. Determine the minimum coefficient μ_s of static friction for which the man's shoes will not slip. Also determine the distance s required for equilibrium of the man's body.



(10.0 Marks)

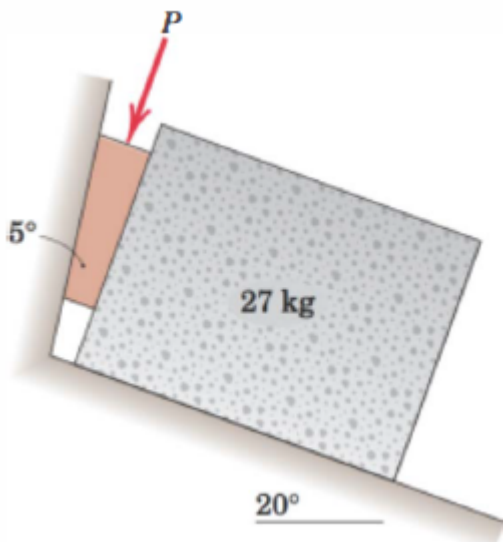
4.c. The two 4° wedges are used to position the vertical column under a load L . What is the least value of the coefficient of friction μ_2 for the bottom pair of surfaces for which the column may be raised by applying a single horizontal force P to the upper wedge?

(5.0 Marks)



4.d. The coefficient of static friction for both wedge surfaces is 0.40 and that between the 27-kg concrete block and the 20° incline is 0.70. Determine the minimum value of the force P required to begin moving the block down the 20° incline as shown. Neglect the weight of the wedge.

(5.0 Marks)





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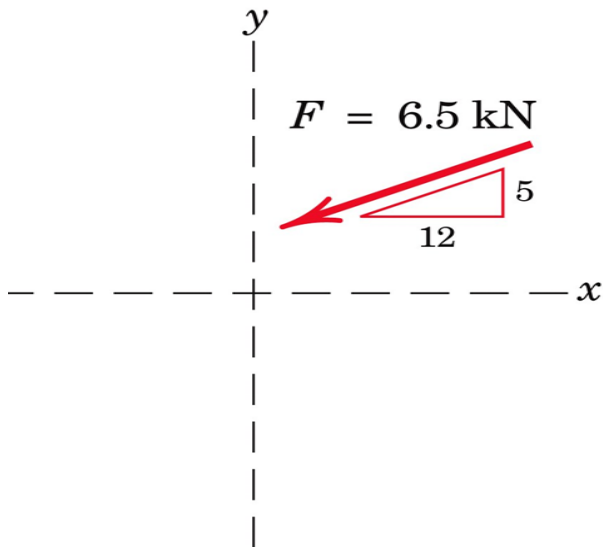
END SEMESTER ASSESSMENT (ESA) - Jan-Feb

UE24CV141A - Engineering Mechanics- Statics

Total Marks : 100.0

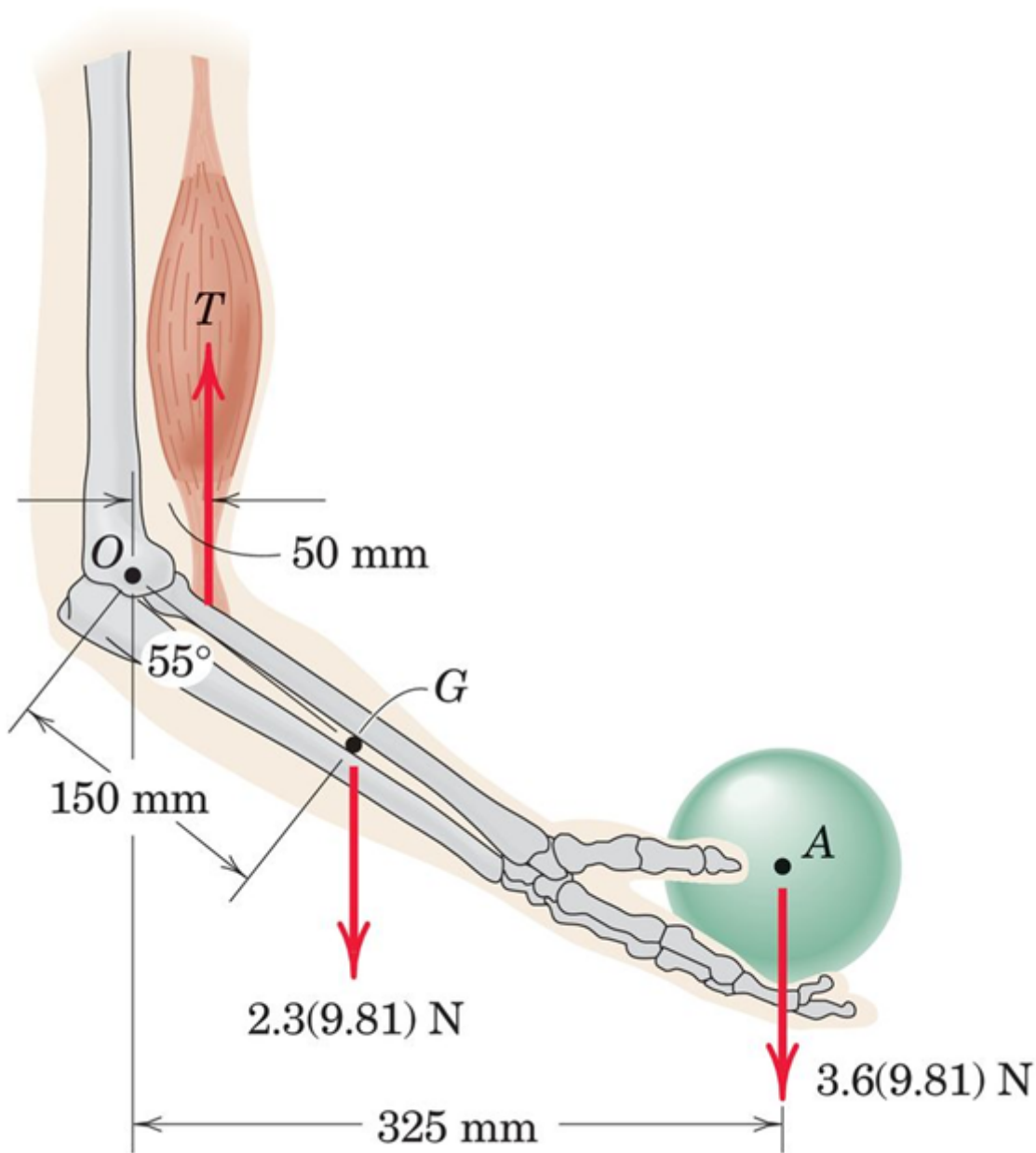
1.a. The slope of the 6.5-kN force F is specified as shown in the figure. Express F as a vector in terms of the unit vectors i and j .

(5.0 Marks)



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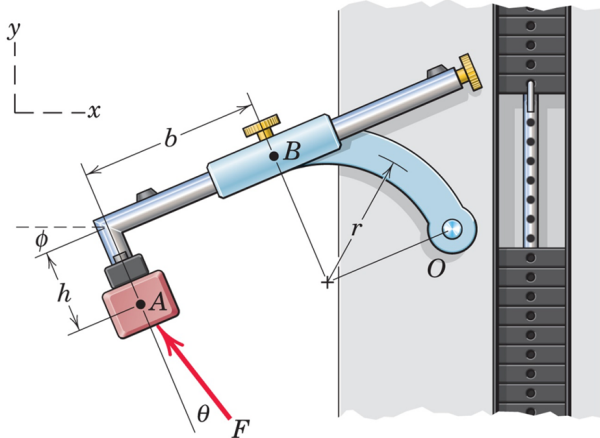
1.b. Elements of the lower arm are shown in the figure. The mass of the forearm is 2.3 kg with center of mass at G. Determine the combined moment about the elbow pivot O of the weights of the forearm and the sphere. What must the biceps tension force be so that the overall moment about O is zero?



(5.0 Marks)

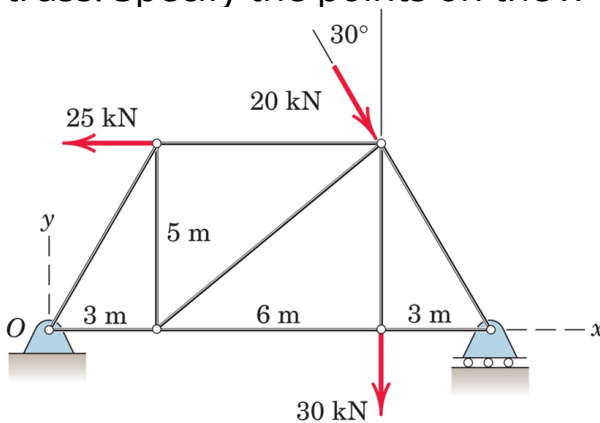
1.c. The force F is applied to the leg-extension exercise machine as shown. Determine the equivalent force couple system at point O . Use values of $F = 520 \text{ N}$, $b = 450 \text{ mm}$, $h = 215 \text{ mm}$, $r = 325 \text{ mm}$, $\theta = 15^\circ$, and $\phi = 100^\circ$.

(6.0 Marks)



1.d. Determine the resultant R of the three forces acting on the simple truss. Specify the points on the x - and y -axes through which R must pass.

(6.0 Marks)

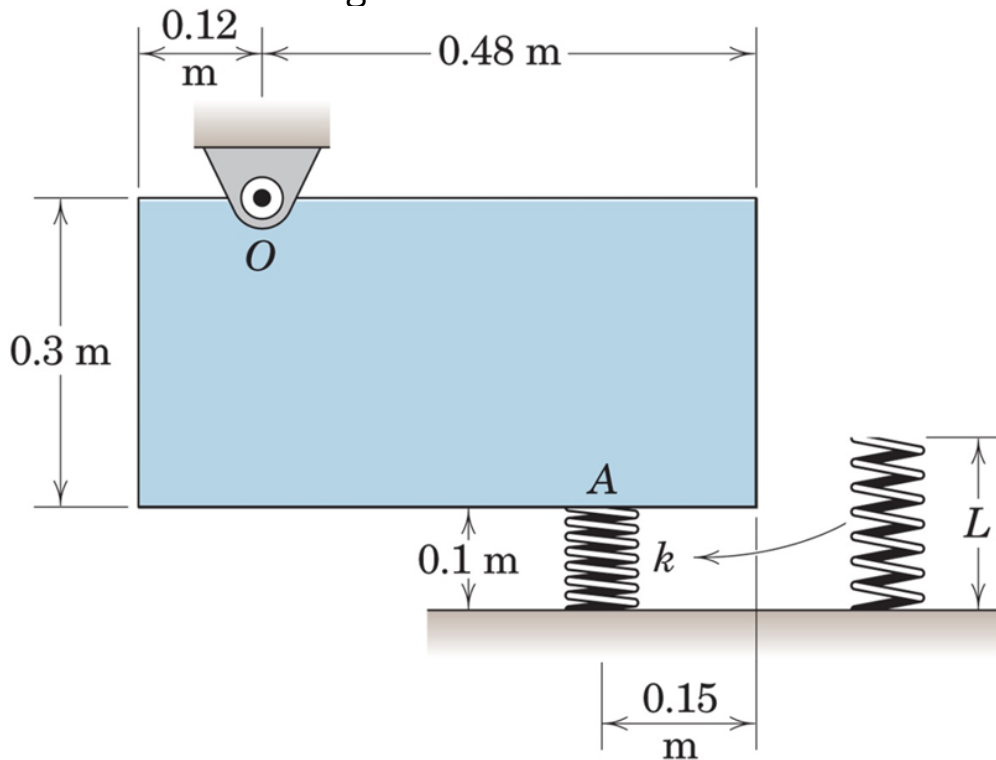


1.e. Define the Principle of transmissibility with neat sketch

(3.0 Marks)

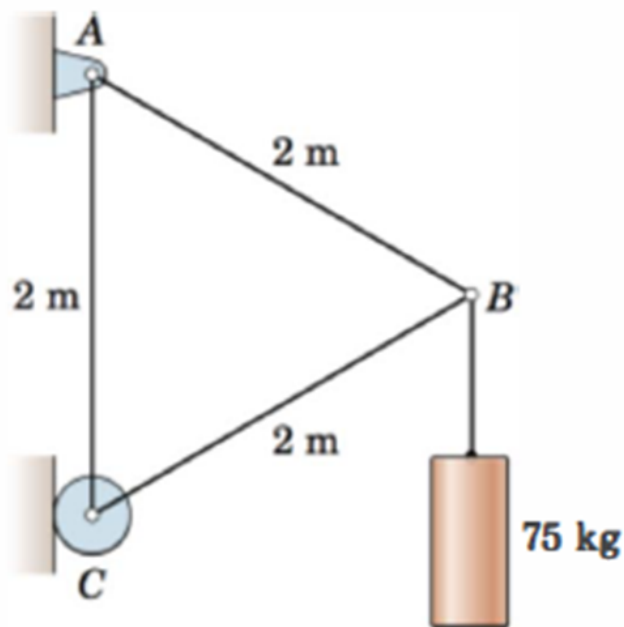
2.a. The 20-kg uniform rectangular plate is supported by an ideal pivot at O and a spring which must be compressed prior to being slipped into place at point A . If the modulus of the spring is $k = 2 \text{ kN/m}$, what must be its undeformed length L ?

(6.0 Marks)



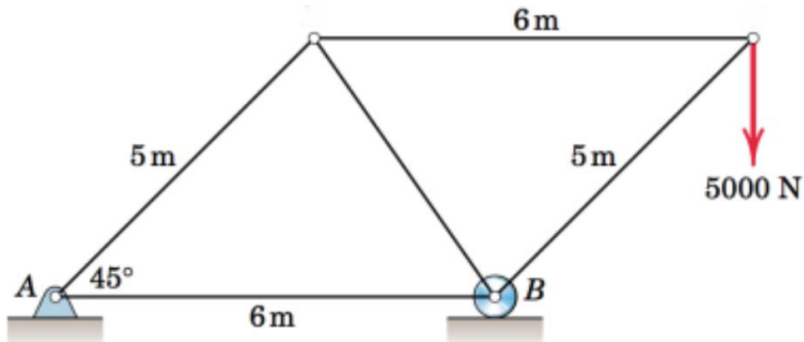
2.b. Determine the force in each member of the simple equilateral truss.

(6.0 Marks)



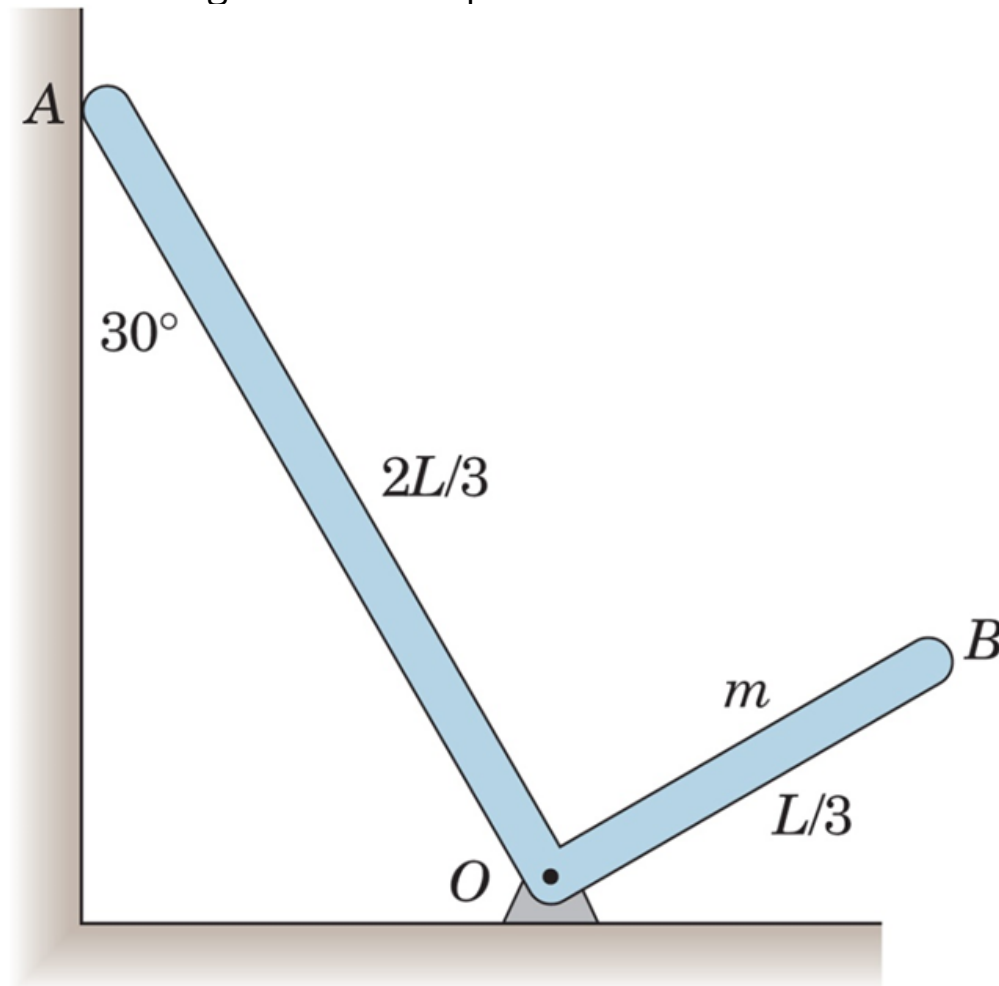
2.c. Determine the force in each member of the loaded truss.

(8.0 Marks)



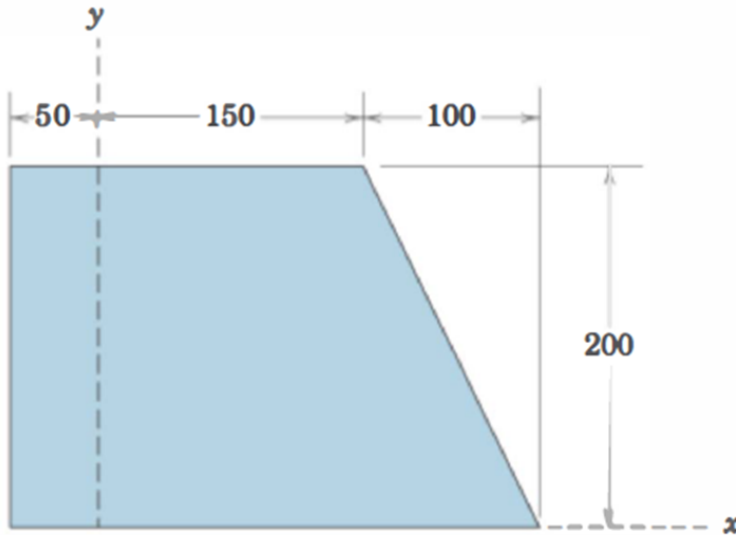
2.d. The right-angle uniform slender bar AOB has mass m . If friction at the pivot O is neglected, determine the magnitude of the normal force at A and the magnitude of the pin reaction at O .

(5.0 Marks)



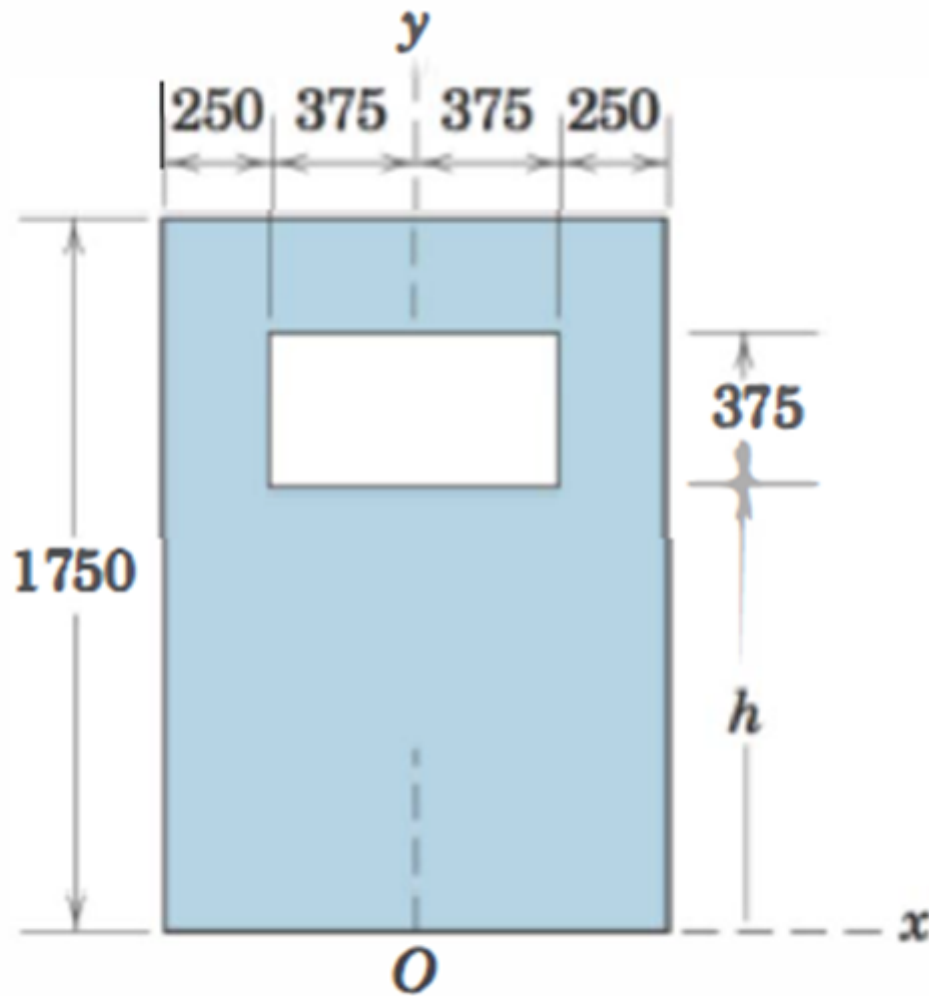
3.a. Determine the x- and y-coordinates of the centroid of the shaded area.

(6.0 Marks)



Dimensions in millimeters

3.b. The variable h designates the arbitrary vertical location of the bottom of the rectangular cutout within the rectangular area. Determine the area moment of inertia about the x -axis for (a) $h = 1000$ mm and (b) $h = 1500$ mm.

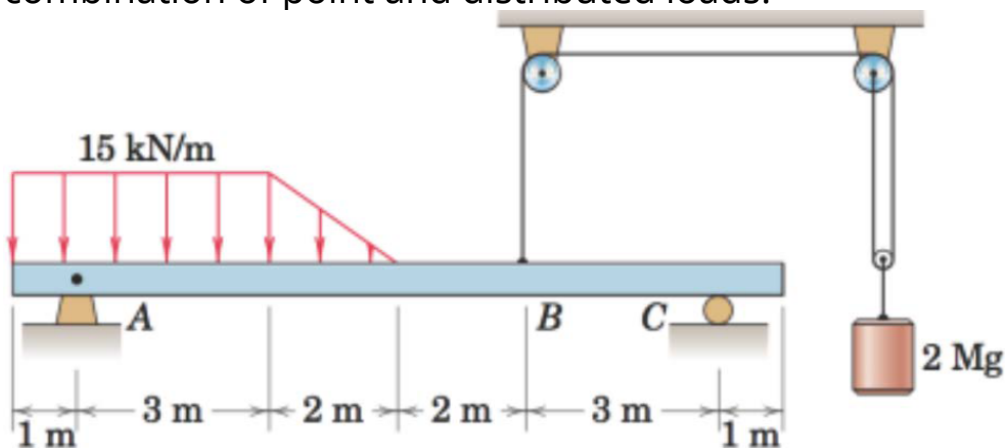


Dimensions in millimeters

(8.0 Marks)

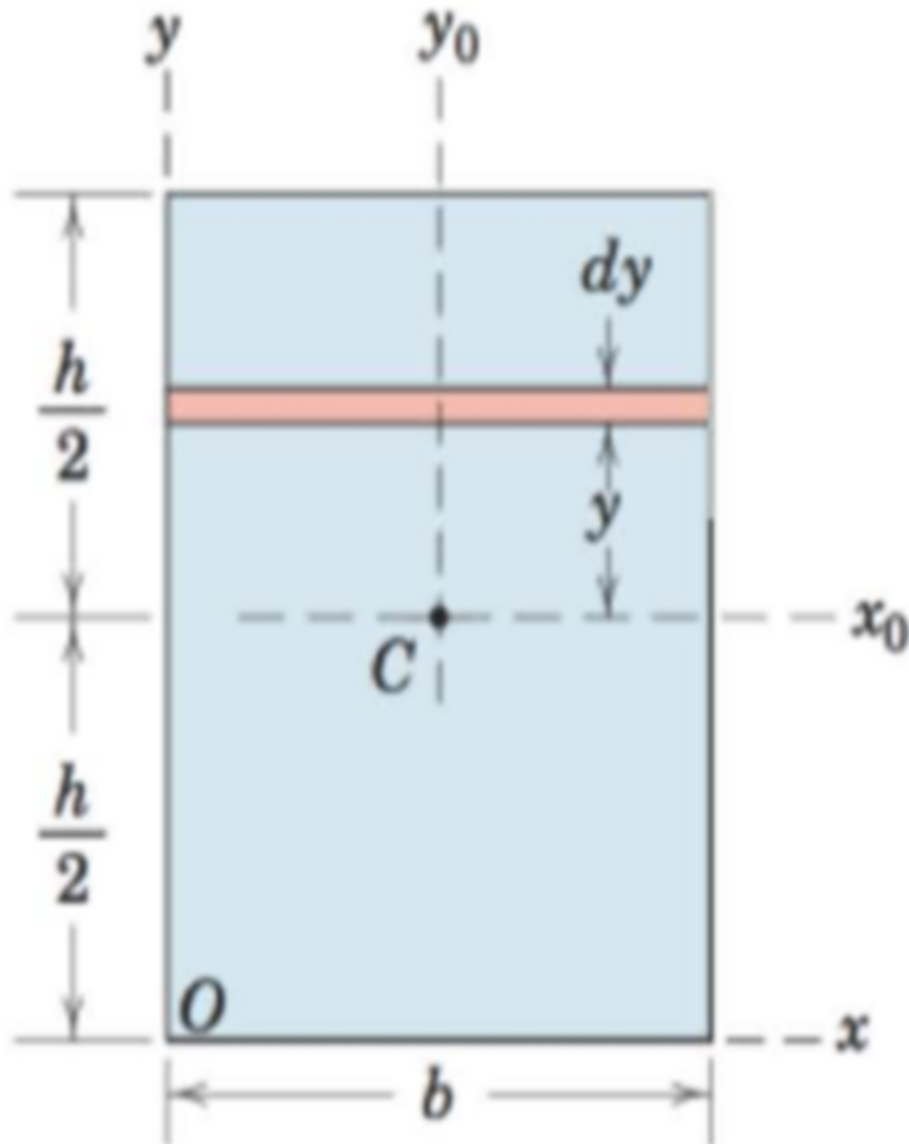
3.c. Determine the reactions at A and C for the beam subjected to the combination of point and distributed loads.

(8.0 Marks)

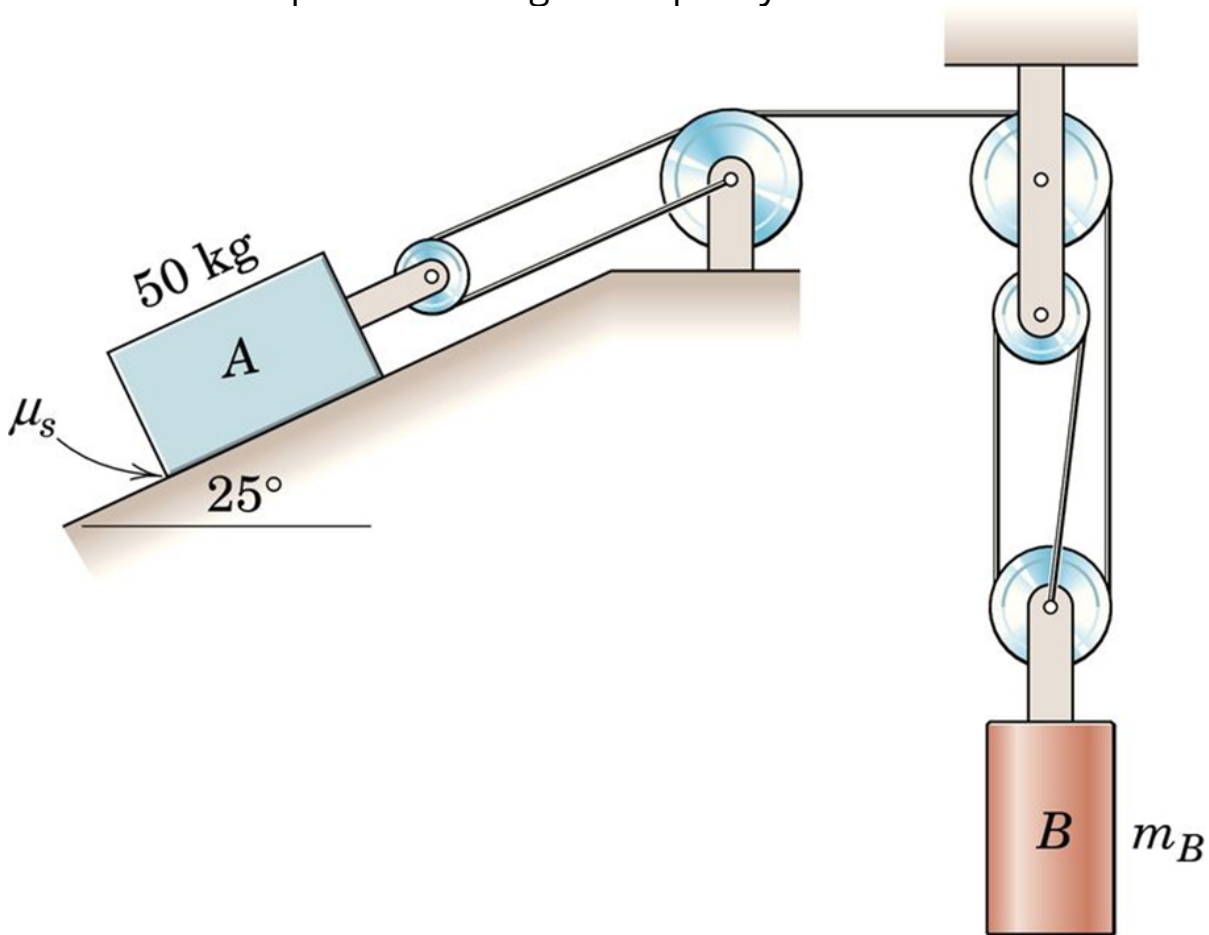


3.d. Derive the expression of Moment of inertia for the rectangular structure

(3.0 Marks)



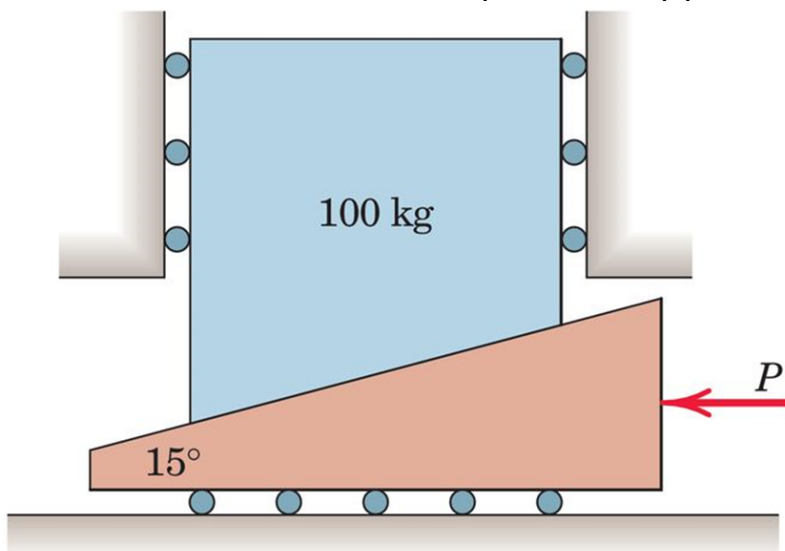
4.a. If the coefficient of static friction between block **A** and the incline is $\mu_s = 0.30$, determine the range of cylinder masses m_B for which the system will remain in equilibrium. Neglect all pulley friction.



(9.0 Marks)

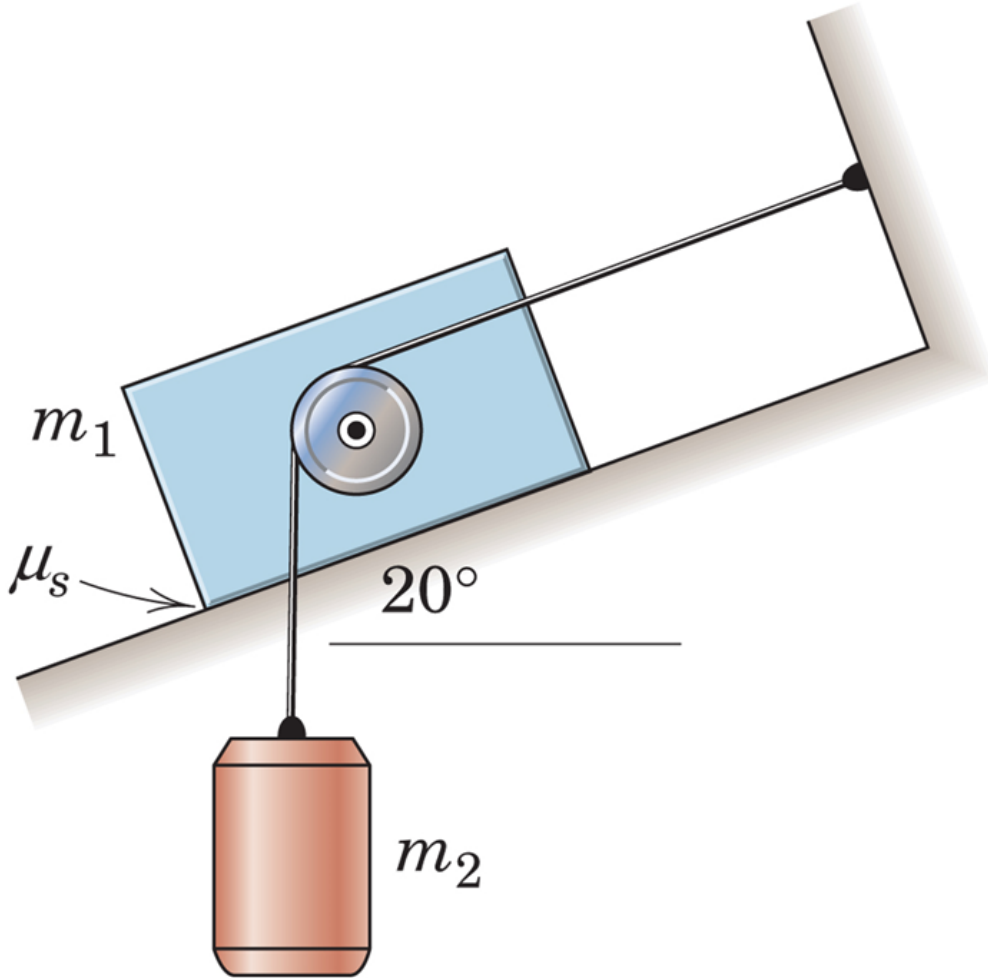
4.b. The coefficient of static friction μ_s between the 100-kg body and the 15° wedge is 0.20. Determine the magnitude of the force P required to begin raising the 100-kg body if (a) rollers of negligible friction are present under the wedge, as illustrated, and (b) the rollers are removed and the coefficient of static friction $\mu_s = 0.20$ applies at this surface as well.

(8.0 Marks)



4.c. Determine the range of mass m_2 for which the system is in equilibrium. The coefficient of static friction between the block and the incline is $\mu_s = 0.25$. Neglect friction associated with the pulley.

(8.0 Marks)





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END SEMESTER ASSESSMENT (ESA) - Feb 2026

UE25CV141A - Engineering Mechanics - Statics

Total Marks : 100.0

1.a. Answer the followings

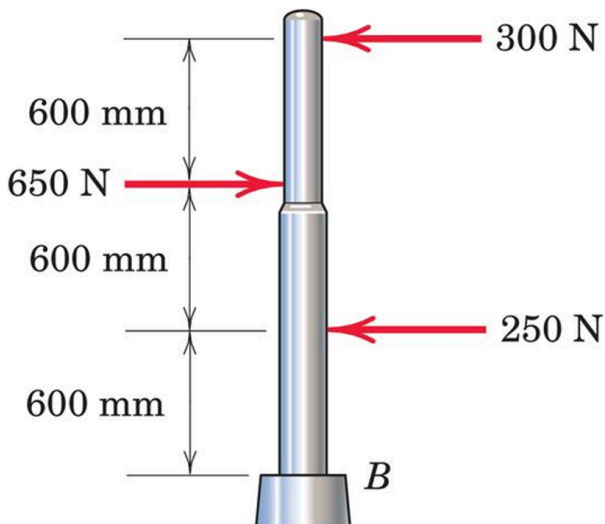
i) Define scalar quantity and vector quantity, give two examples each

ii) Explain with neat sketch Principles of transmissibility

iii) Explain with neat sketch Varignon's theorem (8.0 Marks)

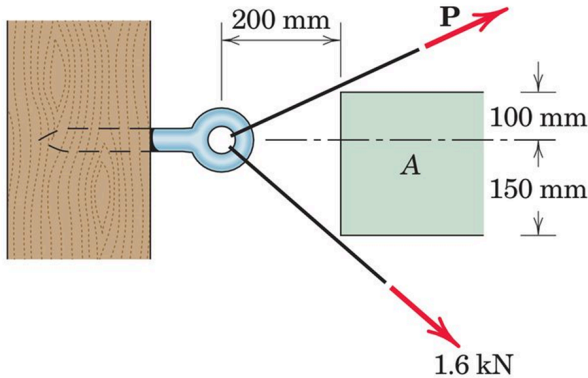
1.b. Determine the height h above the base B at which the resultant of the three forces acts.

(5.0 Marks)



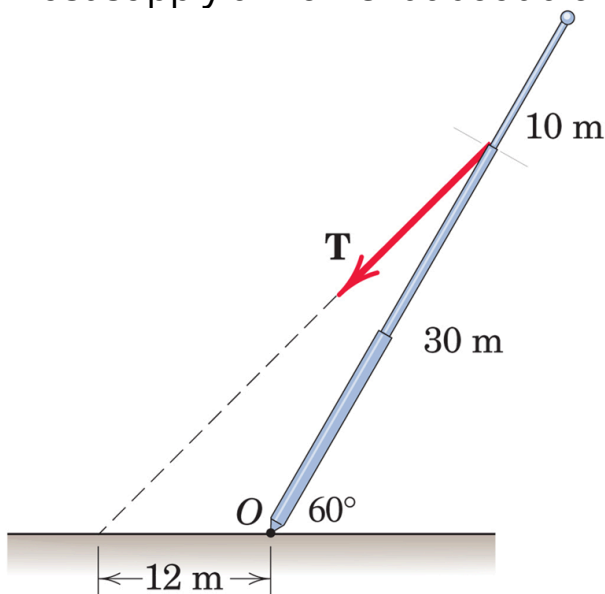
1.c. It is desired to remove the spike from the timber by applying force along its horizontal axis. An obstruction A prevents direct access, so that two forces, one 1.6 kN and the other P, are applied by cables as shown. Compute the magnitude of P necessary to ensure a resultant T directed along the spike. Also find T.

(6.0 Marks)



1.d. In raising the pole from the position shown, the tension T in the cable must supply a moment about O of 72 kN·m. Determine T.

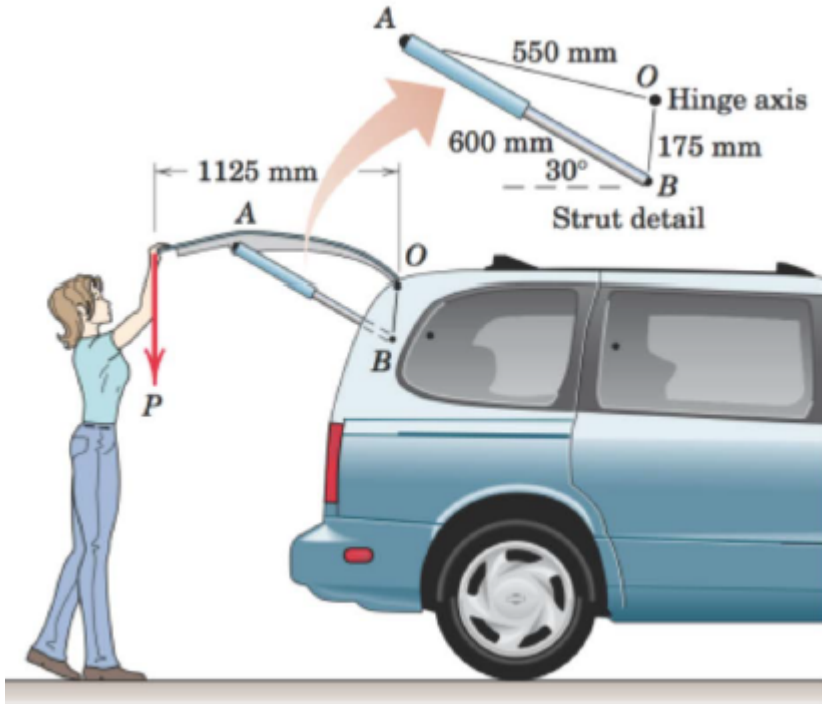
(6.0 Marks)



2.a. What do you understand by the terms 'Roller Support', 'Hinge Support', and 'Fixed Support'? (5.0 Marks)

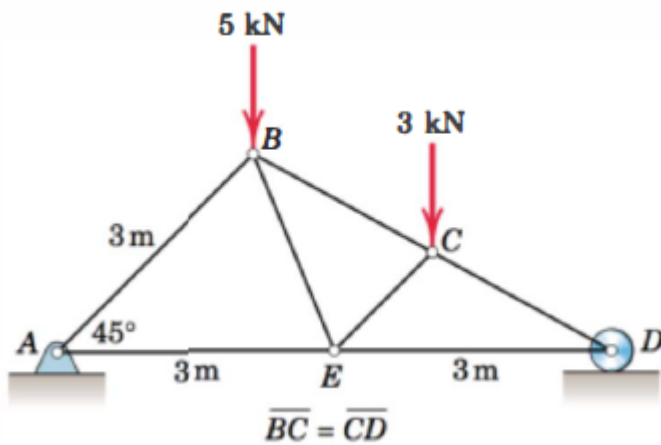
2.b. It is desired that a person be able to begin closing the van hatch from the open position shown with a 40-N vertical force P . As a design exercise, determine the necessary force in each of the two hydraulic struts AB . The mass center of the 40-kg door is 37.5 mm directly below point A . Treat the problem as two-dimensional.

(7.0 Marks)

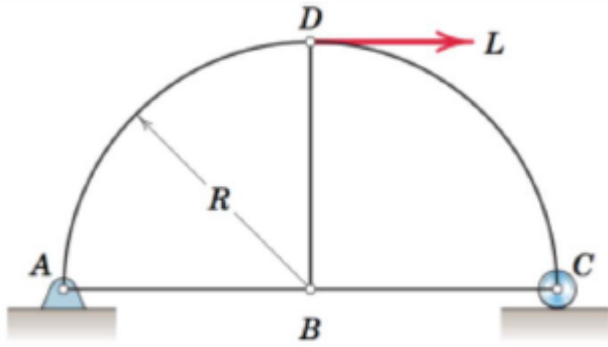


2.c. Determine the forces in members BE and CE of the loaded truss.

(8.0 Marks)

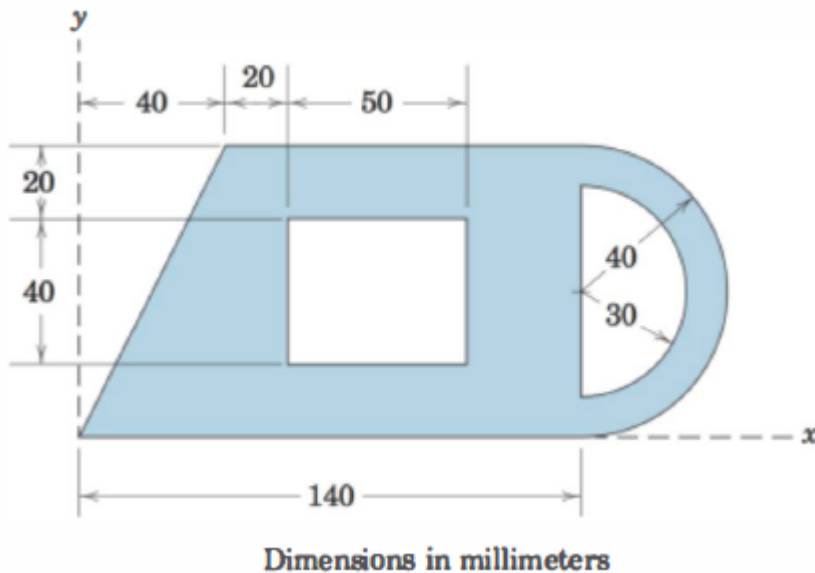


2.d. Determine the forces in members AB, BC, and BD of the loaded truss. (5.0 Marks)



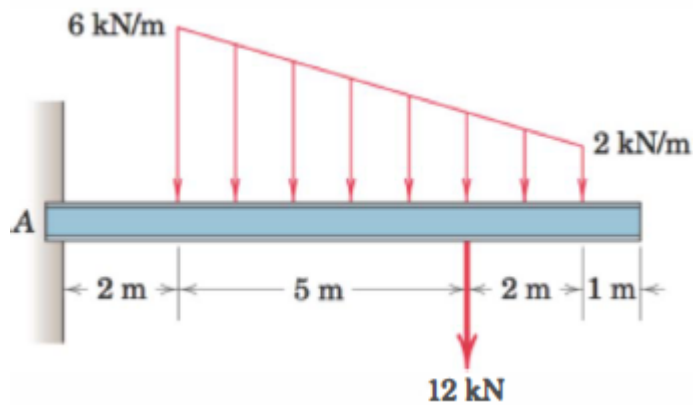
3.a. Determine the Centroid for an “area of a circular sector” (5.0 Marks)

3.b. Determine the coordinates of the centroid of the shaded area. (8.0 Marks)



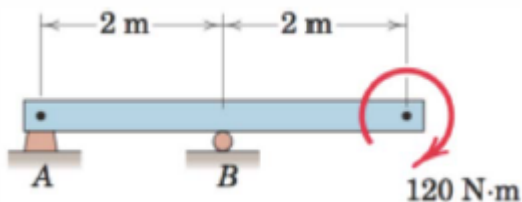
3.c. Determine the force and moment reactions at A for the cantilever beam subjected to the loading shown.

(6.0 Marks)



3.d. Draw the shear and moment diagrams for the beam subjected to the end couple. What is the bending moment M at a section 0.5 m to the right of B?

(6.0 Marks)

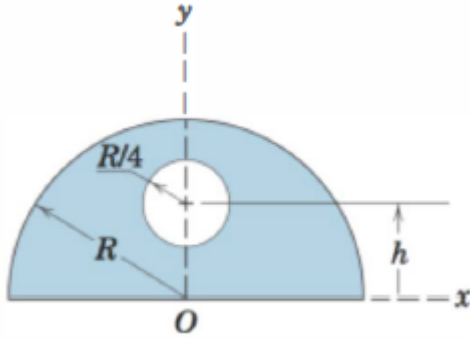


4.a. **State and prove the parallel axis theorem (Transfer of axis).**

(5.0 Marks)

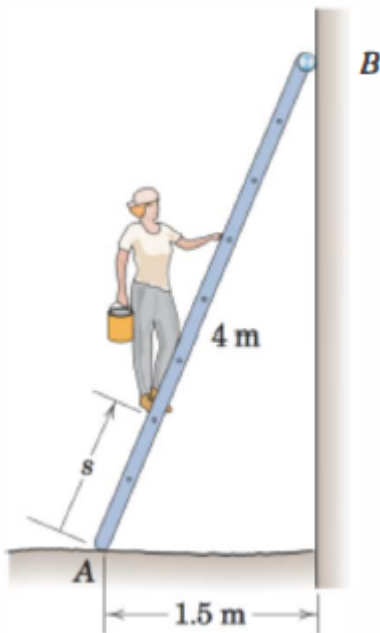
4.b. The variable h designates the arbitrary vertical location of the center of the circular cutout within the semicircular area. Determine the area moment of inertia about the x -axis for (a) $h = 0$ and (b) $h = R/2$.

(7.0 Marks)



4.c. Determine the distances to which the 90-kg painter can climb without causing the 4-m ladder to slip at its lower end A. The top of the 15-kg ladder has a small roller, and at the ground the coefficient of static friction is 0.25. The mass center of the painter is directly above her feet.

(7.0 Marks)



4.d. Determine the minimum coefficient of static friction μ_s which will allow the drum with fixed inner hub to be rolled up the 15° incline at a steady speed without slipping. What are the corresponding values of the force P and the friction force F ?

(6.0 Marks)

